1. Write a recursive C program to find factorial of a given number.

```
#include <stdio.h>

/* Recursive function to find factorial */
int factorial(int n) {
    if (n == 0 || n == 1)
        return 1;
    else
        return n * factorial(n - 1);
}

int main() {
    int n, result;

    printf("Enter a number: ");
    scanf("%d", &n);

    if (n < 0) {
        printf("Factorial not defined for negative numbers\n");
    } else {
        result = factorial(n);
        printf("Factorial of %d = %d\n", n, result);
    }

    return 0;
}
```

Output:

```
Enter a number: 5
Factorial of 5 = 120
```

2. Write a recursive C program to find Fibonacci value of a given number.

```
#include <stdio.h>

/* Recursive function to find Fibonacci */
int fibonacci(int n) {
    if (n == 0)
        return 0;
    else if (n == 1)
        return 1;
    else
        return fibonacci(n - 1) + fibonacci(n - 2);
}
```

```
int main() {
    int n, result;

    printf("Enter the position: ");
    scanf("%d", &n);

    if (n < 0) {
        printf("Fibonacci not defined for negative numbers\n");
    } else {
        result = fibonacci(n);
        printf("Fibonacci value at position %d = %d\n", n, result);
    }

    return 0;
}
```

Output:

```
Enter the position: 7
Fibonacci value at position 7 = 13
```

3.Write a recursive C program to find multiplication of two numbers.

```
#include <stdio.h>

/* Recursive function to find product */
int product(int a, int b) {
    if (b == 1)
        return a;
    else
        return a + product(a, b - 1);
}

int main() {
    int a, b, result;

    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    result = product(a, b);

    printf("Product of %d and %d = %d\n", a, b, result);

    return 0;
```

```
}
```

Output:

Enter two numbers: 6 4
Product of 6 and 4 = 24

4. Write a recursive C program to find GCD of two numbers.

```
#include <stdio.h>

/* Recursive function to find GCD using Euclidean algorithm */
int gcd(int a, int b) {
   if (b == 0)
      return a;
   else
      return gcd(b, a % b);
}

int main() {
   int a, b, result;

   printf("Enter two numbers: ");
   scanf("%d %d", &a, &b);

   result = gcd(a, b);

   printf("GCD of %d and %d = %d\n", a, b, result);

   return 0;
}
```

Output:

Enter two numbers: 48 18
GCD of 48 and 18 = 6

5. Write a recursive C program to solve Towers of Hanoi problem.

```
#include <stdio.h>

/* Recursive function for Towers of Hanoi */
void toh(int n, char source, char destination, char auxiliary) {
```

```
    if (n == 1) {
        printf("Move disk 1 from %c to %c\n", source, destination);
        return;
    }

    toh(n - 1, source, auxiliary, destination);
    printf("Move disk %d from %c to %c\n", n, source, destination);
    toh(n - 1, auxiliary, destination, source);
}

int main() {
    int n;

    printf("Enter number of disks: ");
    scanf("%d", &n);

    printf("Steps to solve Towers of Hanoi:\n");
    toh(n, 'A', 'C', 'B');

    return 0;
}
```

Output:

```
Enter number of disks: 3
Steps to solve Towers of Hanoi:
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 3 from A to C
Move disk 1 from B to A
Move disk 2 from B to C
Move disk 1 from A to C
```